

TEST014 - Environment Configuration Testing

Objective

The purpose of this testing task was to validate that the Food Remedy backend and database environment configuration loads correctly and does not rely on hardcoded sensitive values. This testing also reviewed Firebase credential handling, dotenv configuration, environment variable usage, and local execution of backend/database scripts.

Testing Scope

The following areas were reviewed and tested:

- Cleaning pipeline execution
- Firestore seeding execution (dry-run mode)
- Environment variable handling
- dotenv configuration usage
- Firebase credential configuration
- Sensitive credential protection
- Multi-environment configuration support

Test 1 – Cleaning Pipeline Environment Execution

Description

The database cleaning pipeline was executed locally to verify that the project environment and dependencies were configured correctly.

Result

The cleaning pipeline executed successfully without any configuration-related issues.

Validated:

- Local Python environment setup
- Cleaning pipeline execution
- Database cleaning workflow
- Output file generation

Generated output:

- `database/clean_data/cleanSample.json`

```
Switched to a new branch TEST014-Environment-Configuration-Testing
(base) yanshuhan@yanshuhandeMacBook-Pro Food-Remedy_API % /opt/miniconda3/bin/python3 -m database.clean_data.cleanProductData
WARNING [93552516]: Barcode must be 13 digits
Cleaned data saved to: database/clean_data/cleanSample.json
Total valid products: 11
(base) yanshuhan@yanshuhandeMacBook-Pro Food-Remedy_API %
```

Test 2 – Firestore Seeding Dry-Run Execution

Description

The Firestore seeding engine was executed in dry-run mode to validate environment configuration loading, retry configuration, checkpoint handling, and local execution without performing actual database writes.

Result

The Firestore seeding engine executed successfully.

Validated:

- Firestore configuration loading
- Retry configuration
- Checkpoint handling
- Batch processing
- Dry-run execution
- Local environment execution

Execution Summary:

- Completed: 1
- Failed: 0
- Written: 10

```
(base) yanshuhan@yanshuhandeMacBook-Pro Food-Remedy_API % /opt/miniconda3/bin/python3 database/seeding/seed_firestore.py --dry-run --subset 10
```

```
=====
[SEEDING ENGINE] Enhanced Firestore Seeding with Checkpointing & Retry
=====
Input: database/seeding/products_enriched.json
Output: database/seeding/seeded_products.json
Dry-run: True
Batch size: 500
Target rate: 400 writes/sec
Max retries per batch: 3
[SUBSET] Using first 10 record(s); checkpoint resume disabled.
[Checkpoint] Loaded from /Users/yanshuhan/Food-Remedy_API/database/seeding/checkpoint.json
  Batches: 0 completed, 0 failed
  Documents: 0 written, 0 failed
  Last batch index: 10
[CHECKPOINT] Subset mode: checkpoint resume disabled.
[PROCESSING] Starting batch 1 of 1
```

```
[Progress Bar] 100.0% | Batch 1/1 | 10/10 docs
Written: 10 ✓ | Failed: 0 ✗ | Skipped: 0 ⚪ | Success rate: 100.0%
665762.5 docs/sec | Avg batch: 0.00s | Elapsed: 00:00:00
```

SEEDING SUMMARY

```
=====
Elapsed time: 0.00s
Total batches: 1
  ✓ Completed: 1
  ✗ Failed: 0
Total documents: 10
  ✓ Written: 10
  ✗ Failed: 0
=====
```

Test 3 – Environment Variable Validation

Description

The source code was reviewed to confirm that environment variables are used for configuration loading instead of hardcoded sensitive values.

Validation Performed

Reviewed usage of:

- `os.getenv(...)`
- `os.environ.get(...)`

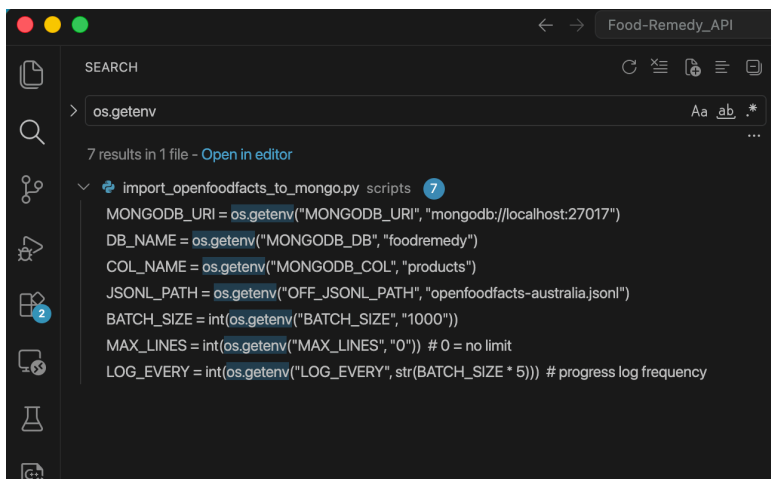
Examples identified:

- `MONGODB_URI`
- `MONGODB_DB`
- `GOOGLE_APPLICATION_CREDENTIALS`
- `FIREBASE_SERVICE_ACCOUNT_KEY`

Result

Environment variables are implemented correctly throughout backend and database configuration handling.

No hardcoded database credentials or Firebase keys were identified during testing.



Test 4 – dotenv Configuration Review

Description

The project configuration structure was reviewed to validate dotenv configuration management.

Validation Performed

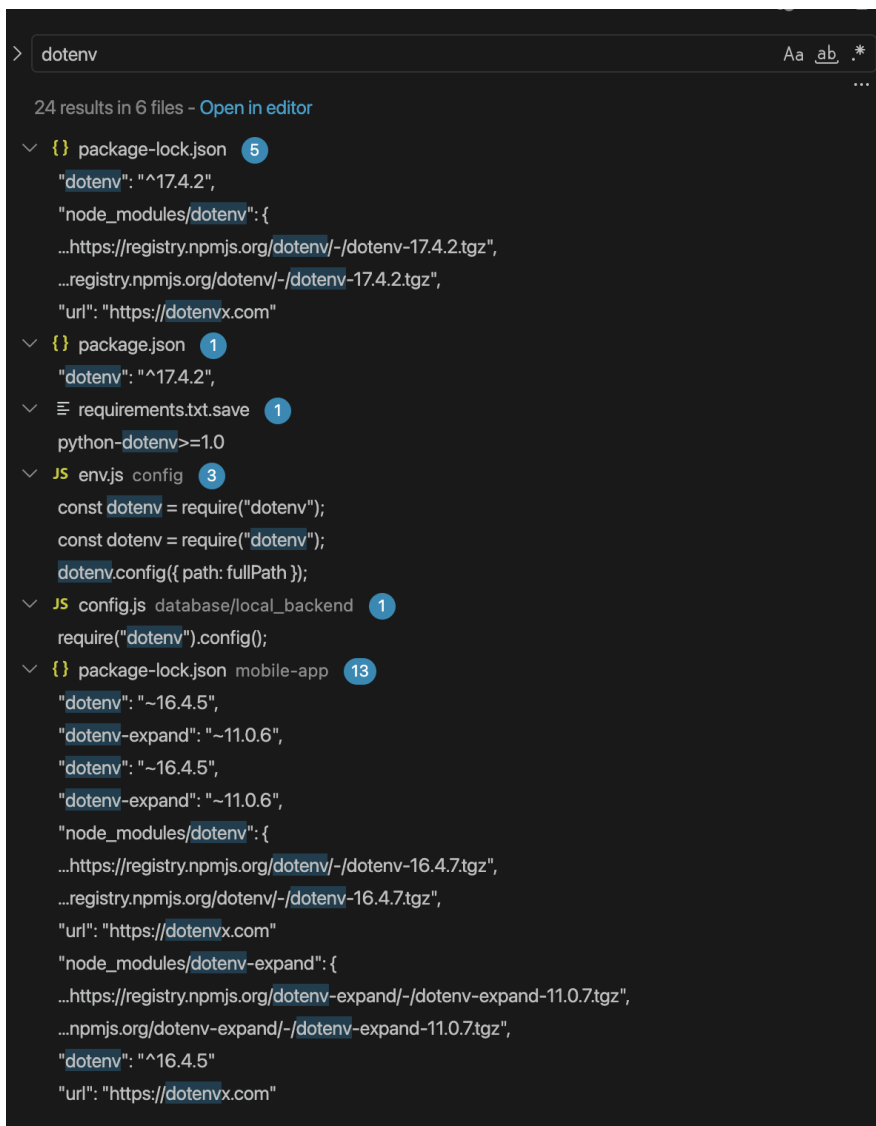
Reviewed:

- `dotenv.config()`
- dotenv package configuration
- Environment loading logic

Result

The project uses dotenv configuration management to separate environment configuration from application logic.

This improves maintainability, deployment flexibility, and configuration security.



The screenshot shows a code editor with a search bar at the top containing the text "dotenv". Below the search bar, it indicates "24 results in 6 files - Open in editor". The search results are listed as follows:

- package-lock.json** (5 results):
 - "dotenv": "^17.4.2",
 - "node_modules/dotenv": {
 - ...https://registry.npmjs.org/dotenv/-/dotenv-17.4.2.tgz",
 - ...registry.npmjs.org/dotenv/-/dotenv-17.4.2.tgz",
 - "url": "https://dotenvx.com"
- package.json** (1 result):
 - "dotenv": "^17.4.2",
- requirements.txt.save** (1 result):
 - python-dotenv>=1.0
- env.js config** (3 results):
 - const dotenv = require("dotenv");
 - const dotenv = require("dotenv");
 - dotenv.config({ path: fullPath });
- config.js database/local_backend** (1 result):
 - require("dotenv").config();
- package-lock.json mobile-app** (13 results):
 - "dotenv": "~16.4.5",
 - "dotenv-expand": "~11.0.6",
 - "dotenv": "~16.4.5",
 - "dotenv-expand": "~11.0.6",
 - "node_modules/dotenv": {
 - ...https://registry.npmjs.org/dotenv/-/dotenv-16.4.7.tgz",
 - ...registry.npmjs.org/dotenv/-/dotenv-16.4.7.tgz",
 - "url": "https://dotenvx.com"
 - "node_modules/dotenv-expand": {
 - ...https://registry.npmjs.org/dotenv-expand/-/dotenv-expand-11.0.7.tgz",
 - ...npmjs.org/dotenv-expand/-/dotenv-expand-11.0.7.tgz",
 - "dotenv": "^16.4.5"
 - "url": "https://dotenvx.com"

Test 5 – Firebase Credential Security Review

Description

Firebase credential handling and configuration were reviewed to validate secure credential management practices.

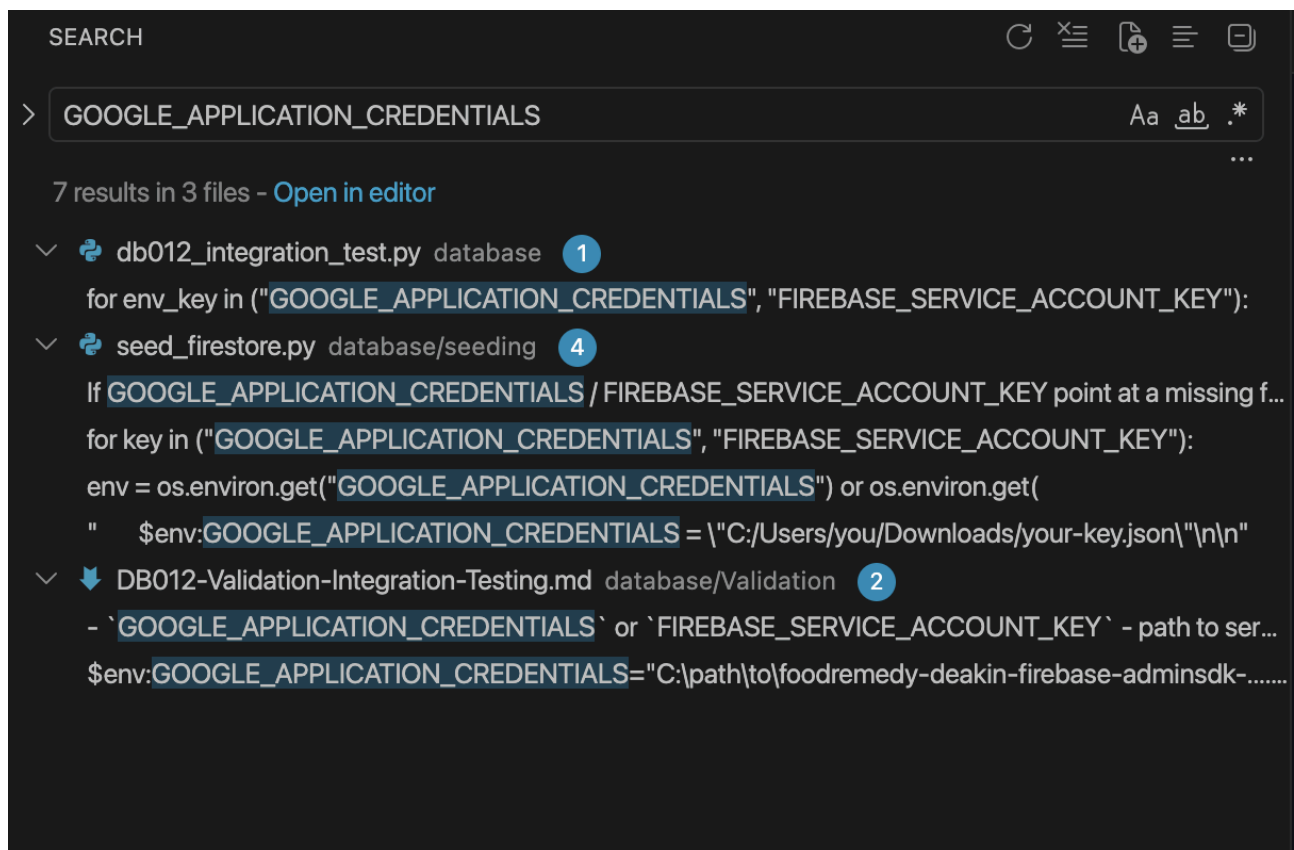
Validation Performed

Reviewed:

- GOOGLE_APPLICATION_CREDENTIALS
- FIREBASE_SERVICE_ACCOUNT_KEY
- Firebase credential loading logic

Result

Firebase credentials are loaded through environment variables and external credential files rather than hardcoded secrets.



The screenshot shows a search interface with the query 'GOOGLE_APPLICATION_CREDENTIALS'. It displays 7 results across 3 files. The results are as follows:

- db012_integration_test.py** (database) - 1 result: A line of code: `for env_key in ("GOOGLE_APPLICATION_CREDENTIALS", "FIREBASE_SERVICE_ACCOUNT_KEY"):`
- seed_firestore.py** (database/seeding) - 4 results: A block of code including a comment: `If GOOGLE_APPLICATION_CREDENTIALS / FIREBASE_SERVICE_ACCOUNT_KEY point at a missing f...`, a loop: `for key in ("GOOGLE_APPLICATION_CREDENTIALS", "FIREBASE_SERVICE_ACCOUNT_KEY"):`, and an environment variable assignment: `env = os.environ.get("GOOGLE_APPLICATION_CREDENTIALS") or os.environ.get(" $env:GOOGLE_APPLICATION_CREDENTIALS = \"C:/Users/you/Downloads/your-key.json\"\\n\\n"`
- DB012-Validation-Integration-Testing.md** (database/Validation) - 2 results: A line of code: `- `GOOGLE_APPLICATION_CREDENTIALS` or `FIREBASE_SERVICE_ACCOUNT_KEY` - path to ser...` and an environment variable assignment: `$env:GOOGLE_APPLICATION_CREDENTIALS="C:\path\to\foodremedy-deakin-firebase-adminsdk-....."`

Test 6 – Sensitive Credential Protection Review

Description

Repository configuration files were reviewed to ensure sensitive Firebase credential files are excluded from Git version control.

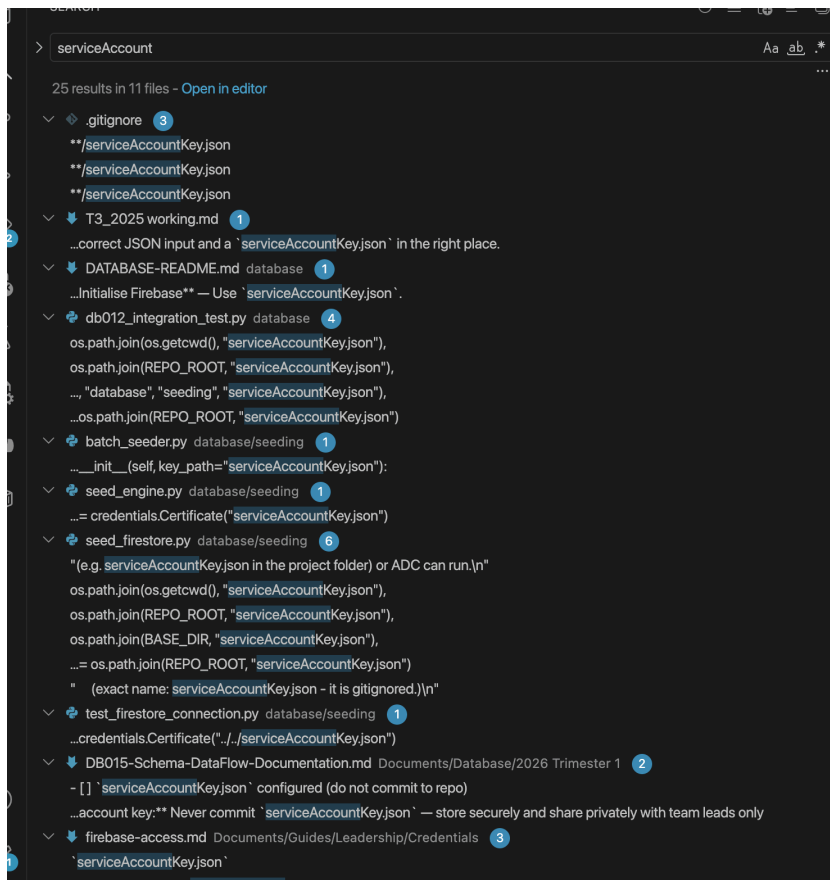
Validation Performed

Reviewed:

- .gitignore
- serviceAccountKey.json

Result

Sensitive Firebase credential files are excluded from Git tracking and are not committed into the repository.



Test 7 – Multi-Environment Configuration Review

Description

The project structure was reviewed to validate support for multiple environment configurations.

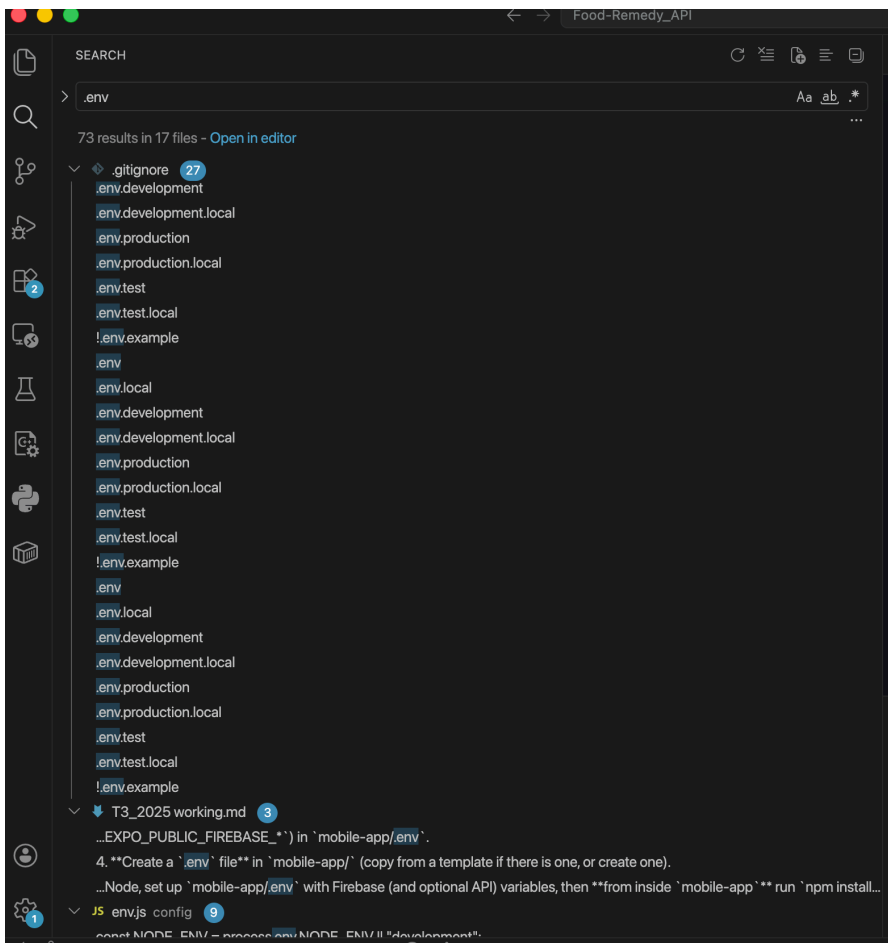
Validation Performed

Reviewed:

- .env
- .env.development
- .env.production
- .env.test
- .env.local

Result

The project supports separate environment configurations for development, testing, and production workflows.



Conclusion

TEST014 – Environment Configuration Testing was completed successfully.

The testing confirmed that:

- Backend and database environment configuration loads correctly
- Cleaning and seeding scripts execute successfully
- Environment variables are implemented properly
- Firebase credentials are handled securely
- Sensitive files are protected from Git tracking
- dotenv configuration is used correctly
- Multiple environment configuration support is available

Overall, the Food Remedy backend and database environment configuration is functioning correctly and follows appropriate configuration management and security practices.